Chapter 12, and the method of FFT convolution in Chapter 18. For now, focus on how signals are convolved by frequency domain multiplication.

To start, we need to define how to multiply one frequency domain signal by another, i.e., what it means to write: $X[f] \times H[f] = Y[f]$. In polar form, the magnitudes are multiplied: $MagY[f] = MagX[f] \times MagH[f]$, and the phases are added: $PhaseY[f] = PhaseX[f] + PhaseH[f]$. To understand this, imagine a cosine wave entering a system with some amplitude and phase. Likewise, the output signal is also a cosine wave with some amplitude and phase. The polar form of the frequency response directly describes how the two amplitudes are related and how the two phases are related.

When frequency domain multiplication is carried out in *rectangular form* there are cross terms between the real and imaginary parts. For example, a sine wave entering the system can produce both cosine and sine waves in the output. To multiply frequency domain signals in rectangular notation:

EQUATION 9-1
Multiplication of frequency domain signals in rectangular form: $Y[f] = X[f] \times H[f]$.

$$ReY[f] = ReX[f]\,ReH[f] - ImX[f]\,ImH[f]$$

$$ImY[f] = ImX[f]\,ReH[f] + ReX[f]\,ImH[f]$$

Focus on understanding multiplication using *polar notation*, and the idea of cosine waves passing through the system. Then simply accept that these more elaborate equations result when the same operations are carried out in rectangular form. For instance, let's look at the *division* of one frequency domain signal by another. In polar form, the division of frequency domain signals is achieved by the inverse operations we used for multiplication. To calculate: $H[f] = Y[f] / X[f]$, divide the magnitudes and subtract the phases, i.e., $MagH[f] = MagY[f] / MagX[f]$, $PhaseH[f] = PhaseY[f] - PhaseX[f]$. In rectangular form this becomes:

EQUATION 9-2
Division of frequency domain signals in rectangular form, where: $H[f] = Y[f] / X[f]$.

$$ReH[f] = \frac{ReY[f]\,ReX[f] + ImY[f]\,ImX[f]}{ReX[f]^2 + ImX[f]^2}$$

$$ImH[f] = \frac{ImY[f]\,ReX[f] - ReY[f]\,ImX[f]}{ReX[f]^2 + ImX[f]^2}$$

Now back to frequency domain convolution. You may have noticed that we cheated slightly in Fig. 9-8. Remember, the convolution of an N point signal with an M point impulse response results in an $N+M-1$ point output signal. We cheated by making the last part of the input signal all *zeros* to allow this expansion to occur. Specifically, (a) contains 453 nonzero samples, and (b) contains 60 nonzero samples. This means the convolution of the two, shown in (c), can fit comfortably in the 512 points provided.